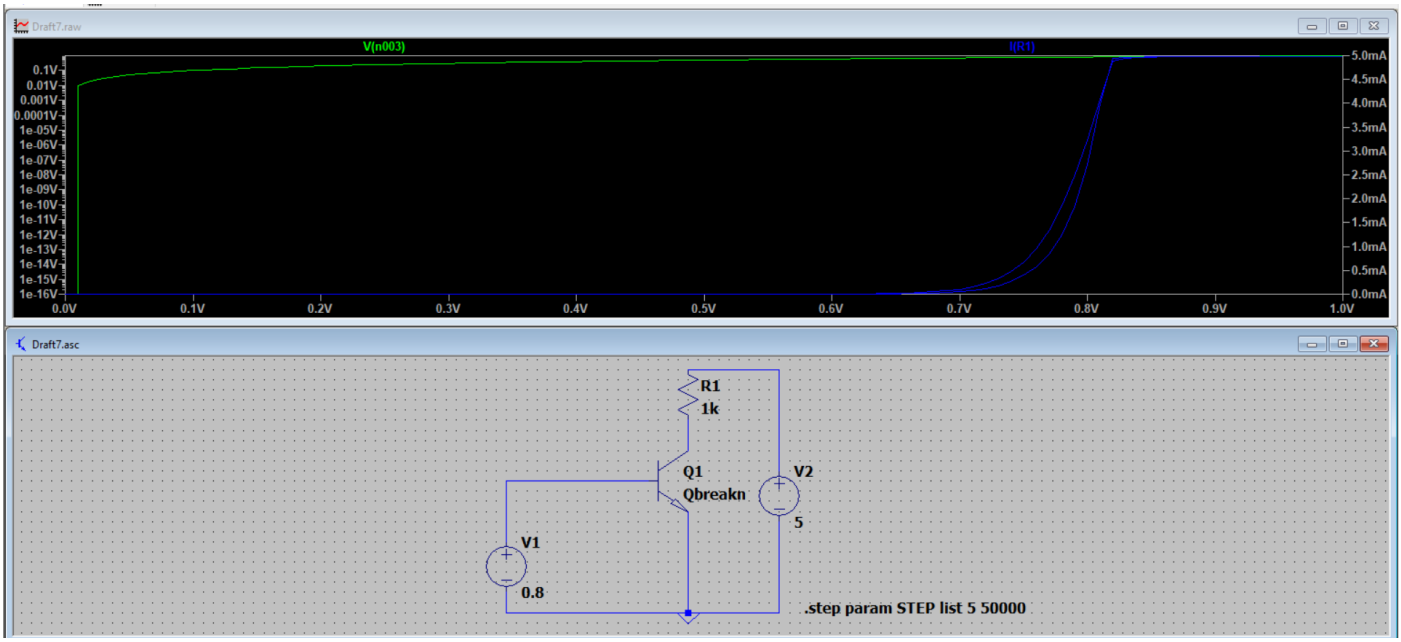
Introdcution

The study of transistor characteristics is fundamental in understanding the behavior and applications of these essential electronic components. In this laboratory session, we explore Bipolar Junction Transistors (BJTs). The primary objective of this experiment is to gain insights into transistor characteristics, specifically focusing on the measurement of current gain and obtaining hands-on experience with transistor circuits.

2- Preliminary Work

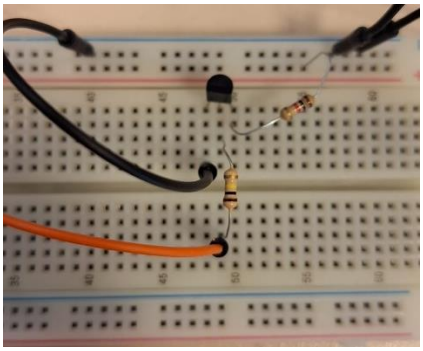


3- Lab Tasks

The laboratory tasks are designed to provide practical experience with 2N4401 (NPN) and 2N4403 (PNP) BJT transistors.

3.1 – Task 1

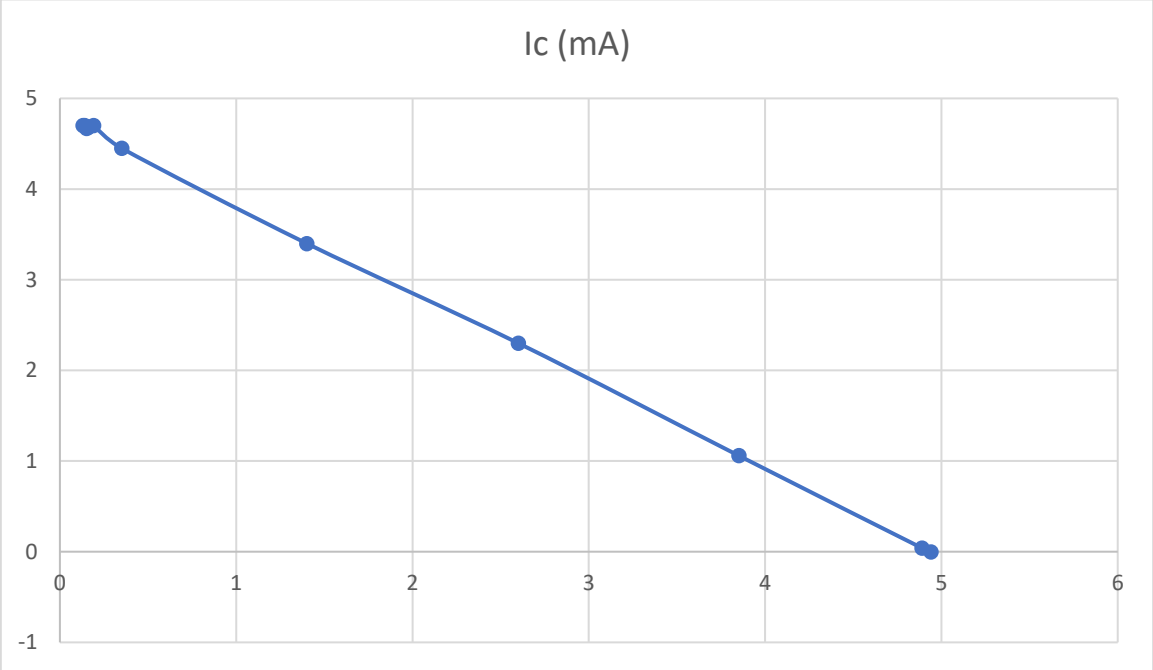
- First, we have asked to construct cir 3.1 with 2N4401 transistor, +5V fixed as an input. Then we connected V_{BB} to variable voltage source.



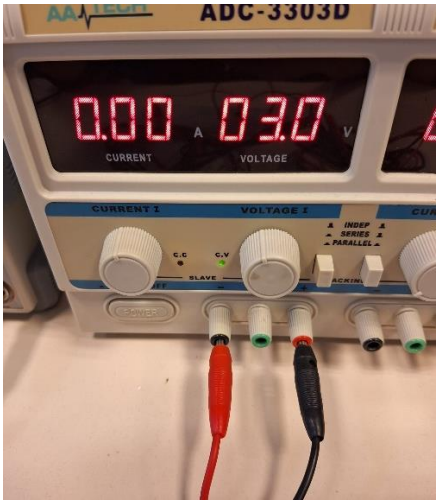
- Then we have varied V_{BB} from 0V to 5V with 0.5V steps ans measured voltage drops over the base and collector sides of transistor. After measuring voltages of V_{100k} V_{BE} I_B V_{1k} V_{BC} I_C V_{CE} we have calculated Beta of the transistor for each measurment.

VBB (V)	V100k (V)	VBE (V)	IB (mA)	V1k (V)	VBC (V)	Ic (mA)	VCE (V)	β
0	0.001	0.005	0.00001	0	-4.89	0	4.94	0
0.5	0.015	0.54	0.00015	0.04	-4.27	0.04	4.89	266.6667
1	0.42	0.62	0.0042	1.06	-3.15	1.06	3.85	252.381
1.5	0.91	0.65	0.0091	2.3	-2	2.3	2.6	252.7473
2	1.3	0.66	0.013	3.4	-1	3.4	1.4	261.5385
2.5	1.7	0.66	0.017	4.45	-0.3	4.45	0.35	261.7647
3	2.3	0.66	0.023	4.7	0.46	4.7	0.19	204.3478
3.5	2.85	0.66	0.0285	4.68	0.5	4.68	0.16	164.2105
4	3.3	0.66	0.033	4.67	0.51	4.67	0.15	141.5152
4.5	3.82	0.66	0.0382	4.7	0.52	4.7	0.14	123.0366
5	4.3	0.66	0.043	4.7	0.53	4.7	0.13	109.3023

- We have found Beta value of transistor with $\beta = I_C / I_B$ equation.



- On the next step of the task 1, we have to examine temperature effect on the collector current. To do that experiment we have heat up the transistor, we simply heat transistor with holding it. Before heating up the transistor we get 1.06V with 1kΩ resistor, which gives us 1.06mAmp current on the collector.
- After heating up the transistor our measurments were got up to 1.12V with 1kΩ resistor, which is 1.12mAmp current on the collector. Which is 1.06 times higher before heating the transistor. Heating up is increases number of free electrons so we get more current passes through.
- Then we set the V_{BB} to +4V and V_{CC} to 2V and set V_{BB} to -3V and V_{CC} to 5V measured V_{100k} V_{BE} I_B V_{1k} V_{BC} I_C V_{CE}. (To measure the -3V on V_{BB} we simply reversed the cables on the power supply.)



Vcc	VBB (V)	V100k (V)	VBE (V)	IB (mA)	V1k (V)	VBC (V)	Ic (mA)	VCE (V)	B
2	4	3.38	0.65	0.0338	1.88	-0.53	1.88	0.11	55.6213
5	-3	0	3	0	0	-8	0	5	-

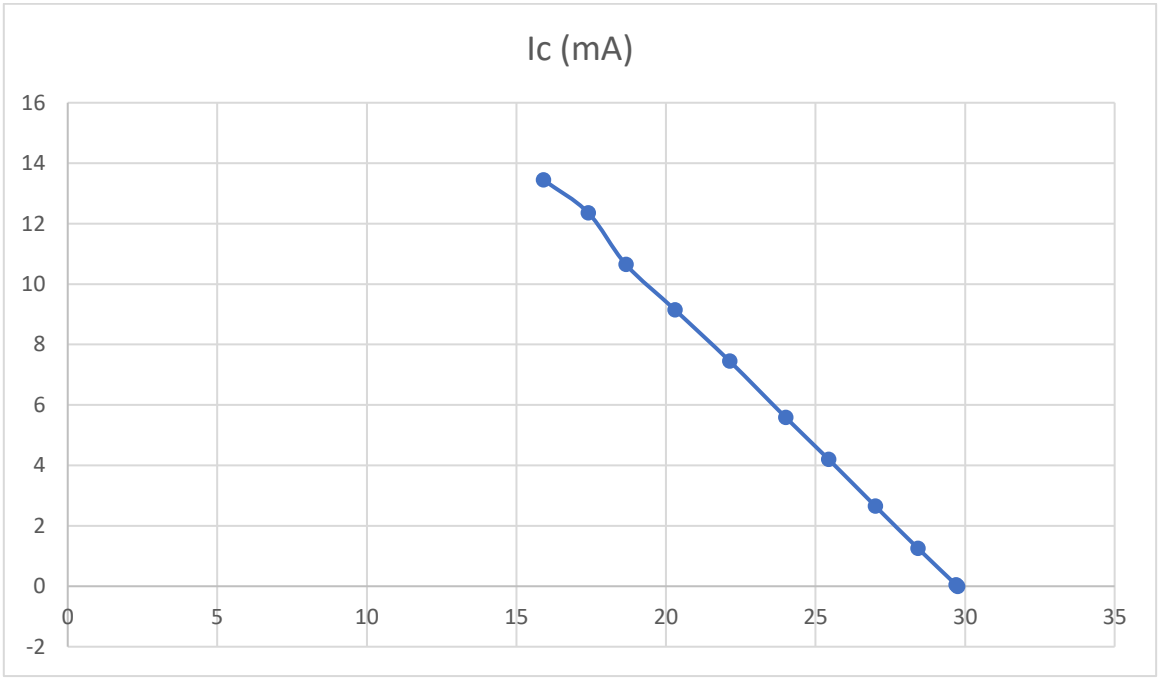
- We have nothing on the 5 V_{CC} with -3 V_{BB} because we did not have any current.
- And for Vcc = 2V and V_{BB} = 4V is on the active region and Vcc = 5V and V_{BB} = -3V is it on the

3.2 – Task 2

- For the task 2 we repeat the first 5 steps of Task 1 with 30V supply voltage to plot IC vs VCE

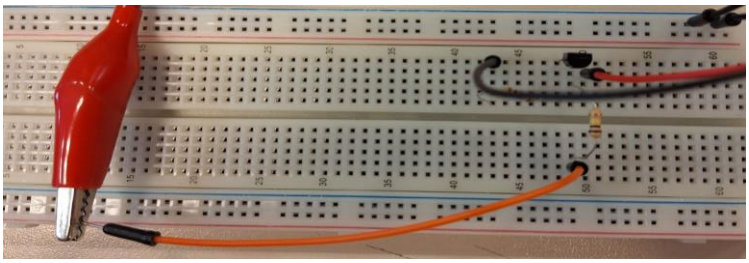


VBB (V)	V100k (V)	VBE (V)	IB (mA)	V1k (V)	VBC (V)	Ic (mA)	VCE (V)	B
0	0	0.005	0	0	-29.5	0	29.75	-
0.5	0.02	0.54	0.0002	0.048	-28.5	0.048	29.7	240
1	0.42	0.62	0.0042	1.26	-27	1.26	28.42	300
1.5	0.88	0.63	0.0088	2.65	-25.7	2.65	27	301.1364
2	1.31	0.63	0.0131	4.2	-24	4.2	25.44	320.6107
2.5	1.78	0.63	0.0178	5.59	-22.56	5.59	24	314.0449
3	2.4	0.62	0.024	7.45	-20.87	7.45	22.13	310.4167
3.5	2.73	0.618	0.0273	9.15	-19.45	9.15	20.3	335.1648
4	3.32	0.617	0.0332	10.65	-18	10.65	18.66	320.7831
4.5	3.85	0.615	0.0385	12.36	-17.56	12.36	17.4	321.039
5	4.22	0.62	0.0422	13.45	-16.9	13.45	15.9	318.7204



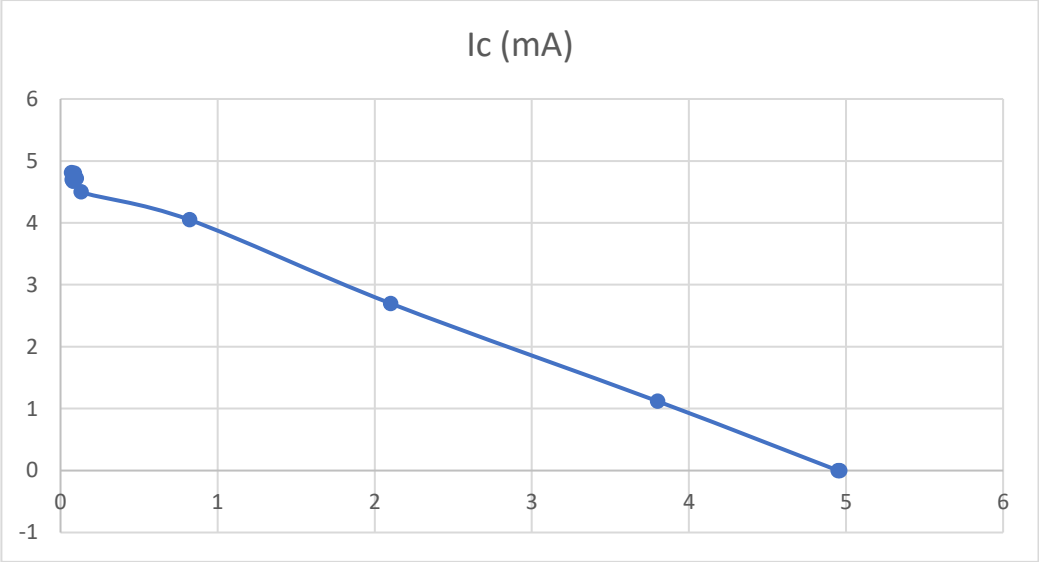
3.1 – Task 2

- Firstly we constructed the Cir 3.2 with 2N4403 transistor, +5V fixed input voltage, connected V_{BB} to the variable voltage source.



- Then we vary VBB from 0 to 5V with 0.5V steps, measure voltage drops over the base and collector resistors and calculate corresponding currents and V_{100k} V_{BE} I_B V_{1k} V_{BC} I_C V_{CE} and β.

VBB (V)	V100k (V)	VEB (V)	IB (mA)	V1k (V)	VCB (V)	Ic (mA)	VEC (V)	B
0	4.27	0.66	0.0427	4.809	0.6	4.809	0.07	112.623
0.5	3.67	0.66	0.0367	4.7	0.58	4.7	0.074	128.0654
1	3.2	0.66	0.032	4.67	0.58	4.67	0.08	145.9375
1.5	2.72	0.65	0.0272	4.8	0.57	4.8	0.088	176.4706
2	2.23	0.64	0.0223	4.72	0.55	4.72	0.1	211.6592
2.5	1.7	0.63	0.017	4.5	0.52	4.5	0.13	264.7059
3	1.25	0.61	0.0125	4.05	0.17	4.05	0.82	324
3.5	0.8	0.6	0.008	2.7	1.48	2.7	2.1	337.5
4	0.3	0.59	0.003	1.12	3.1	1.12	3.8	373.3333
4.5	0	0.41	0	0	4.45	0	4.95	-
5	0	0.122	0	0	4.94	0	4.96	-



3- Conclusion

To sum up, our exploration into Diode Circuits shed light on how diodes behave and their role in rectification. The close match between what we observed and what theory predicts tells us that diode principles like forward bias, reverse bias, and rectification efficiency hold true. Armed with this practical know-how, we're now better equipped to use diodes in electronic circuits, handling concepts like impedance matching and signal rectification with confidence. This hands-on experience adds depth to our understanding of diode behavior, paving the way for more effective use in a variety of electronic applications. Because of some errors caused by the imperfect environment we might had some difference between hands-on measurments and calculations.